Bihar Engineering University, Patna

B.Tech. 7th Semester Examination, 2023

Course: B.Tech. *Time: 03 Hours*

Code:110703 *Subject: High Voltage Engineering* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

(ii) There are NINE questions in this paper.

(iii) Attempt FIVE questions in all.

(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct answer of the following (Any seven question only):*** [2 x 7 = 14]

(a) For the high voltage conductors at high pressures, if the voltage is positive then the corona appears as a ____
(i) Uniform bluish white sheath (ii) Reddish glowing spots
(iii) Uniform greenish spots (iv) None of these

(b) A gas in normal state is almost a perfect
(i) Conductor (ii) Insulator
(iii) Semi-conductor (iv) Dielectric

(c) Which of the following method or technique can be used for the measurement of high dc voltages?
(i) Generating voltmeter (ii) Electrostatic voltmeter
(iii) Peak voltmeter (iv) Any of the above

(d) For voltage measurement, the gap which gives highest accuracy is
(i) Sphere gaps (ii) Field gaps
(iii) Rod gaps (iv) None of these

(e) The sparkover voltage
(i) Increase with humidity
(ii) decrease with the partial pressure of water vapour in air
(iii) Humidity effect decreases with the size of sphere
(iv) Humidity is minimum for uniform field gaps

(f) The commonly used materials for HV insulation is
(i) Paper-oil insulation (ii) Polyester resin
(iii) Ceramic (iv) PVC

(g) For very small cavities, discharge inception voltage
(i) Decreases as the cavity depth increases
(ii) Increases as the cavity depth increases
(iii) Remains unaffected as the cavity depth increases
(iv) None of these

(h) Treeing phenomenon is observed in
(i) Capacitors (ii) Cables
(iii) Insulators (iv) Only (a) and (b)

(i) Liquids dielectrics are mainly used as
(i) Impregnants in high voltage cables (ii) In capacitors
(iii) For filling up transformers (iv) All of these

(j) Cables are tested for high voltage using _____
(i) power frequency voltage (ii) ac-dc voltage
(iii) impulse voltage (iv) power frequency, ac-dc and impulse voltage

Q.2 (a) Explain Thermal breakdown in solid dielectrics. [7]
(b) How this Thermal breakdown is more significant than the other mechanisms? [7]

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Q.3 (a) With a neat sketch, explain the construction and working of a Van de Graff generator. [7]

(b) Why is controlled tripping necessary in impulse generators? Discuss how is it performed using Trigatron gap? [7]

Q.4 Explain in details about the procedure for conducting dielectric, impulse voltage and short circuit current tests on high voltage circuit breakers. [14]

Q.5 (a) Describe the various factors that influence breakdown in gas. [7]

(b) What is composite insulation? How does short-term breakdown differ from long-term breakdown? [7]

Q.6 A 100 kVA, 250 V/200 kV feed transformer has resistance and reactance of 1% and 5% respectively. This transformer is used to test a cable at 400 kV at 50 Hz. The cable takes a charging current of 0.5 A at 400 kV. Determine the series inductance required. Assume 1% resistance of the inductor. Also determine input voltage to the transformer. Neglect dielectric loss of the cable. [14]

Q.7 Explain the terms [14]

(a) withstand voltage,

(b) flashover voltage

(c) 50% flashover voltage, and

(d) wet and dry power frequency tests as referred to high voltage testing.

Q.8 An impulse generator has eight stages with each condenser rated for 0.16 μF and 125 kV. The load capacitor available is 1000 pF. Find the series resistance and the damping resistance needed to produce 1.2/50 μs impulse wave. What is the maximum output voltage of the generator, if the charging voltage is 120 kV? [14]

Q.9 (a) How do the temperature and moisture effect the breakdown strength of solid dielectrics? [7]

(b) Give the temperature classification for solid insulating materials. Why is this classification not done for liquids and gases? [7]

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